VARUN SREENIVASAN

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EXPERIENCE

DataChat, Inc., Madison, WI **Feb 2022 - Present**

Senior Software Engineer

- *Scalability Upgrades:* Proposed & implemented Artillery driven load tests, fixed bottlenecks, and boosted platform functionality.
- *Led a team of developers* to overhaul & integrate VectorDB platform with critical providers Google Vector Search & Milvus.
- *Increased Automation & Delivery Cadence:* Led CI/CD push to streamline release mgmt. process involving Kubernetes deployments.
- *Accelerated LLM adoption:* Set up evaluation framework for benchmarking LLMs across vital use cases; led research assessing feasibility of hosting DataChat's custom LLMs across multiple cloud providers and platforms.
- *Expanded GenAI & ML Product Capabilities:* Spearheaded design & implementation of backend solutions; achieved scalability, reliability and compute efficiency, utilizing system design best practices and cloud native architecture.
- Consistently received high ratings across all performance appraisals & evaluations.

National Science Foundation – IRIS HEP, Berkeley, CA **May 2021 - August 2021**

Fellow – Graph Methods for Particle Tracking (High Luminosity Large Hadron Collider)

- *Optimized Particle Differentiation:* Developed embedding pipeline to find distance metric between 3D hit measurements pairs.
- *Accelerated Nearest Neighbors Search:* Replaced Facebook's Faiss with Fixed Radius Nearest Neighbors (FRNN) on CUDA.

Citrine Informatics, Redwood City, CA **May 2018 - August 2018**

NextGen-Fellow – Computational Materials Science

- *Multi-University Research Project:* Competitively selected, successfully completed bootcamp at Stanford University, presented at NextGen Research Symposium in Golden, CO, and *co-authored a paper*.
- *Optimized Metal Defect Detection:* Trained RetinaNet and developed evaluation pipeline; achieved 85% precision & 68% recall.

EDUCATION

M.S. Computer Science, University of Wisconsin-Madison *(Dec 2021)*

G.P.A: ***4.00/4.00***

B.S. Computer Science (Minor: Mathematics), University of Wisconsin-Madison *(May 2020)*

G.P.A: ***3.90/4.00*** *"Distinction in the Major"* and *Dean's List*

LANGUAGES & TECHNOLOGY SKILLSETS

Python, Git, AWS, GCP, Azure, Terraform, Docker, Kubernetes, Helm, CircleCI, VS Code, Java, Jupyter Notebook, Unix, MacOS

LICENSES & CERTIFICATIONS

Amazon Web Services: *Solutions Architect Associate*; *Cloud Practitioner*; *Machine Learning Specialty*

OTHER SOFTWARE R&D PROJECTS

Autonomous Vehicles

Live Traffic-Sign Detection: Trained PyTorch MobileNet detection model; developed proof of concept for open source testbed; *presented at ACM MobiArch'22*.

Instance Segmentation App

Segmentation With User Images: Created app that accurately identifies and separates objects of interest from background.

Business Success & Viability Forecast

Developed Multiple ML models with Yelp Dataset: Architected feature engineering, created training and validation pipelines, and determined vital factors to predict Covid period business survival & success.

RELEVANT COURSEWORK

Machine Learning, Data Science, Artificial Intelligence, Algorithms, Computer Networks, Mobile & Wireless Networks, Computer Vision, Cryptography, Combinatorics, Spatial Web & Mobile Programming, Linear Algebra & Differential Equations, Operating Systems.

PUBLICATIONS & PRESENTATIONS

- µCity: a miniatured autonomous vehicle testbed: *Procs. 17th ACM Workshop* ***MobiArch'22***
- Multi defect detection & analysis of electron microscopy images with deep learning: Science Direct (Computational Material Sci.)
- Graph methods for particle tracking: IRIS HEP fellowship final presentation